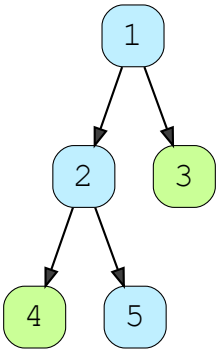
Solution: Level Order Successor

Problem Statement

Given a binary tree and a node, find the level order successor of the given node in the tree. The level order successor is the node that appears right after the given node in the level order traversal.

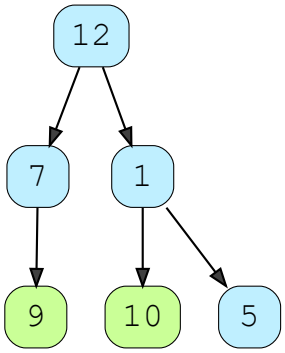
Example 1:

Given Node: 3
Level Order Successor: 4



Example 2:

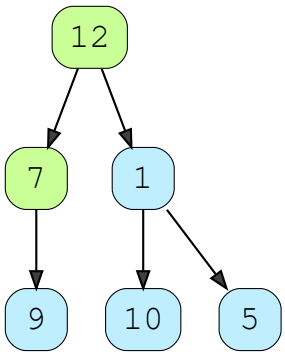
Given Node: 9
Level Order Successor: 10



Example 3:

Given Node: 12

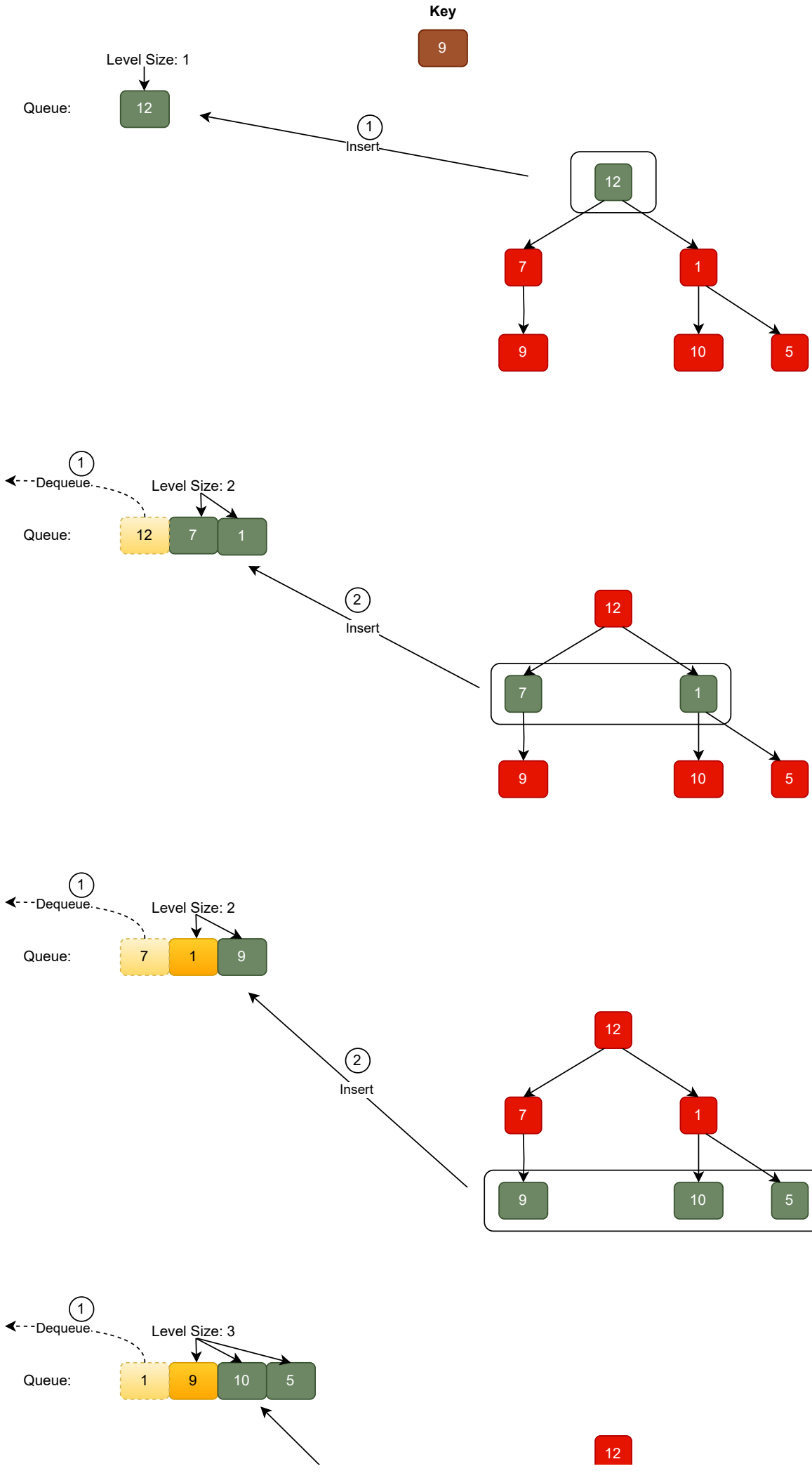
Level Order Successor: 7

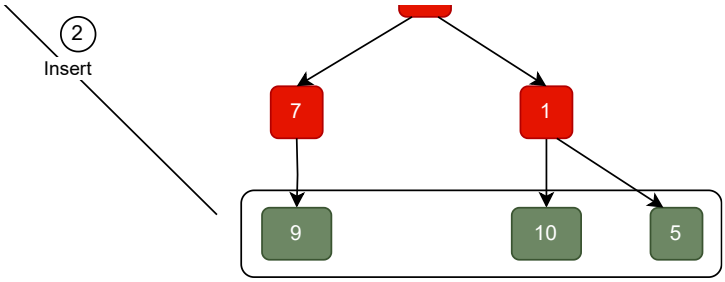


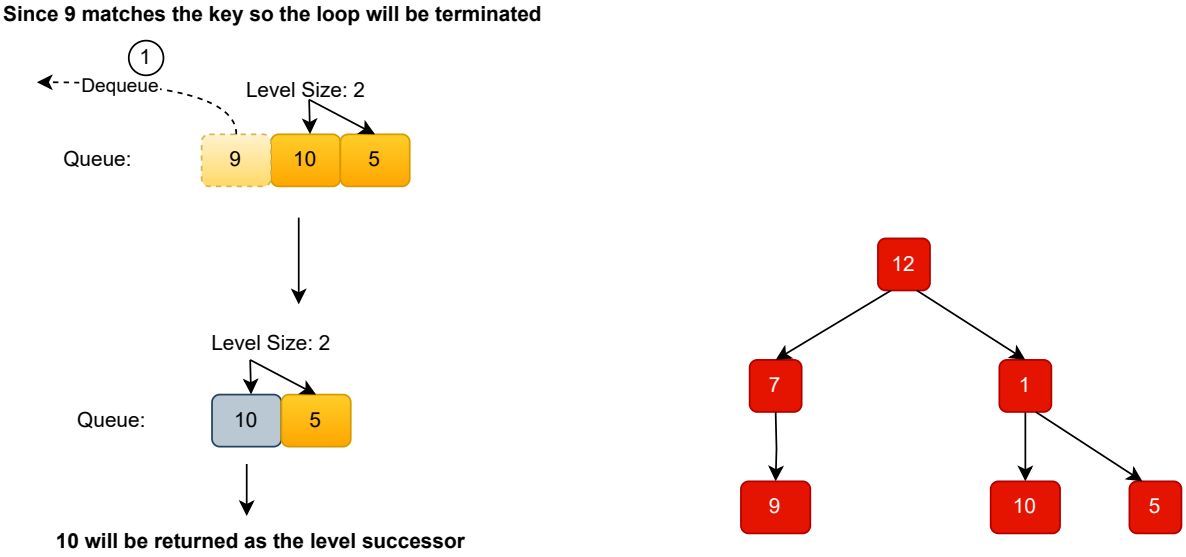
Solution

This problem follows the **Binary Tree Level Order Traversal** pattern. We can follow the same **BFS** approach. The only difference will be that we will not keep track of all the levels. Instead we will keep inserting child nodes to the queue. As soon as we find the given node, we will return the next node from the queue as the level order successor.

Here is the visual representation of the algorithm:


Key
9
Level Size: 1
Queue:
12
1
Insert
12
7
1
9
10
5
1
Dequeue
Level Size: 2
Queue:
12
7
1
2
Insert
12
7
1
9
10
5
1
Dequeue
Level Size: 2
Queue:
7
1
9
2
Insert
12
7
1
9
10
5
1
Dequeue
Level Size: 3
Queue:
1
9
10
5
12





Level Order Successor

Code

Here is the code for this algorithm:

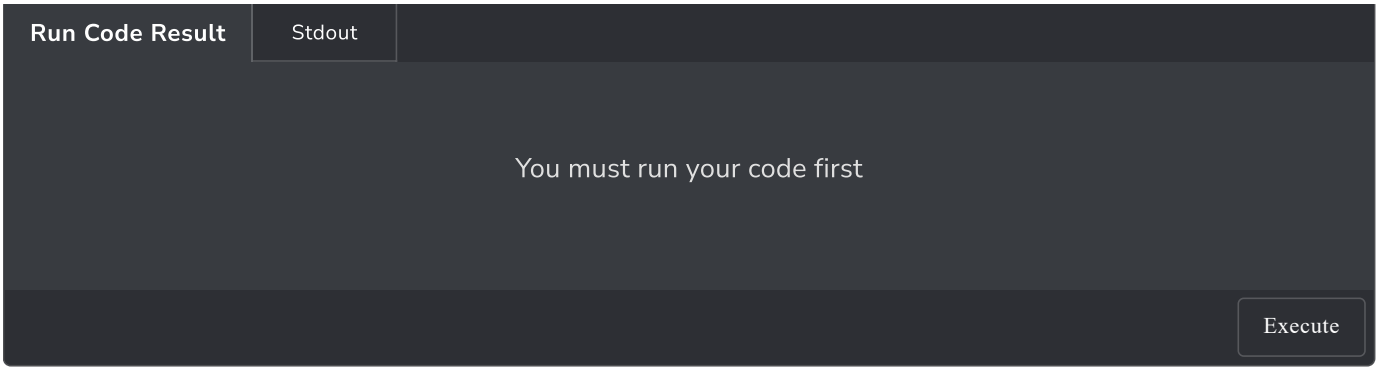
```java
import java.util.*;

// class TreeNode {
//   int val;
//   TreeNode left;
//   TreeNode right;

//   TreeNode(int x) {
//     val = x;
//   }
// };

class Solution {
  public TreeNode findSuccessor(TreeNode root, int key) {
    if (root == null)
      return null;

    Queue<TreeNode> queue = new LinkedList<>();
    queue.offer(root);
    while (!queue.isEmpty()) {
      TreeNode currentNode = queue.poll();
      // insert the children of current node in the queue
      if (currentNode.left != null)
        queue.offer(currentNode.left);
```



Time Complexity

The time complexity of the above algorithm is O(N), where 'N' is the total number of nodes in the tree. This is due to the fact that we traverse each node once.

Space Complexity

The space complexity of the above algorithm will be O(N) which is required for the queue. Since we can have a maximum of N/2 nodes at any level (this could happen only at the lowest level), therefore we will need O(N) space to store them in the queue.

← Previous

Level Order Successor (Easy)

Next →

Connect Level Order Siblings (Medium)

Mark as Completed